Addendum to “A lower bound on the relative L^1-norm of Poincaré series”: Wolfram Mathematica code for computing the integers $N_{n,m,l}^{(p)}$ and $N_{m,l}^{(p)}$

Sonja Žunar Kožić
University of Zagreb, Faculty of Geodesy

In this addendum, we provide a Wolfram Mathematica 14.3 code that computes the integers $N_{n,m,l}^{(p)}$ of Corollary 4.3 for $n \in \{1, 2\}$, as well as a code that computes the integers $N_{m,l}^{(p)}$ of Corollary 5.3.

1. A binary search algorithm for finding the smallest integer $N \geq m$ with a given property

Given a strictly positive integer m, the function `smallestN` below returns the smallest integer $N \geq m$ such that f holds.

```
mid[low_, high_] := If[high == Infinity, 2 * low, Floor[(low + high) / 2]]
binarySearch[f_, low_, high_] := If[low == high, high, If[f[mid[low, high]],
binarySearch[f, low, mid[low, high]], binarySearch[f, mid[low, high] + 1, high]]]
smallestN[f_, m_] := binarySearch[f, m, Infinity]
```

2. Computing the integers $N_{1,m,l}^{(p)}$

Given $p \in [0, 1[$, the code below constructs and prints a table `Ntable1` of integers $N_{1,m,l}^{(p)}$ for $l = 0, 1, \ldots, l_{\max}$ and $m = 3, 4, \ldots, m_{\max}$. The three occurrences of the symbol ⌘ in the code should be replaced with the actual values of p, $l_{\max}$ and $m_{\max}$.

```
phi1[x_, m_, l_] := x^(l / 2) * (1 - x)^(m / 2 - 2);
integral[m_, l_] := Integrate[phi1[x, m, l], {x, 0, 1}];
integraln[m_, l_, n_] :=
  Integrate[phi1[x, m, l], {x, 0, 1 / (Sqrt[1 + 4 / n^2] + 2 / n)^2}];
p = ⌘;
mmax = ⌘;
lmax = ⌘;
Ntable1 = Table[0, {mmax - 2}, {lmax + 1}];
Do[rhs = (p + 1) / 2 * integral[m, l];
Ntable1[[m - 2, l + 1]] = smallestN[Re[integraln[m, l, #]] > rhs &, 3],
{m, 3, mmax}, {l, 0, lmax}];
Print[TableForm[Ntable1]];
```

2.1. Computing the integers $N^{(0)}_{1,m,l}$

The code below constructs and prints a table `Ntable1` of integers $N^{(0)}_{1,m,l}$ for $m = 3, 4, \ldots, 12$ and $l = 0, 1, \ldots, 10$.

```
phi1[x_, m_, l_] := x^(l / 2) * (1 - x)^(m / 2 - 2);
integral[m_, l_] := Integrate[phi1[x, m, l], {x, 0, 1}];
integraln[m_, l_, n_] :=
  Integrate[phi1[x, m, l], {x, 0, 1 / (Sqrt[1 + 4 / n^2] + 2 / n)^2}];
p = 0;
mmax = 12;
lmax = 10;
Ntable1 = Table[0, {mmax - 2}, {lmax + 1}];
Do[rhs = (p + 1) / 2 * integral[m, l];
Ntable1[[m - 2, l + 1]] = smallestN[Re[integraln[m, l, #]] > rhs &, 3],
{m, 3, mmax}, {l, 0, lmax}];
Print[TableForm[Ntable1]];
```

```
14    23    32    40    49    58    67    75    84    93    102
6     9     12    15    18    21    24    26    29    32    35
4     6     8     10    11    13    15    16    18    20    22
4     5     6     7     9     10    11    12    13    15    16
3     4     5     6     7     8     9     10    11    12    13
3     4     5     5     6     7     8     8     9     10    11
3     3     4     5     5     6     7     7     8     9     9
3     3     4     4     5     6     6     7     7     8     8
3     3     4     4     5     5     6     6     7     7     8
3     3     3     4     4     5     5     6     6     7     7
```

Table 2.1. The computed table `Ntable1` of integers $N^{(0)}_{1,m,l}$ for $m = 3, 4, \ldots, 12$ and $l = 0, 1, \ldots, 10$

$m \backslash l$	0	1	2	3	4	5	6	7	8	9	10
3	14	23	32	40	49	58	67	75	84	93	102
4	6	9	12	15	18	21	24	26	29	32	35
5	4	6	8	10	11	13	15	16	18	20	22
6	4	5	6	7	9	10	11	12	13	15	16
7	3	4	5	6	7	8	9	10	11	12	13
8	3	4	5	5	6	7	8	8	9	10	11
9	3	3	4	5	5	6	7	7	8	9	9
10	3	3	4	4	5	6	6	7	7	8	8
11	3	3	4	4	5	5	6	6	7	7	8
12	3	3	3	4	4	5	5	6	6	7	7

2.2. Computing the integers $N_{1,m,l}^{(0.5)}$

The following code constructs and prints a table `Ntable1` of integers $N_{1,m,l}^{(0.5)}$ for $m = 3, 4, \ldots, 12$ and $l = 0, 1, \ldots, 10$.

```
phi1[x_, m_, l_] := x^(l / 2) * (1 - x)^(m / 2 - 2);
integral[m_, l_] := Integrate[phi1[x, m, l], {x, 0, 1}];
integraln[m_, l_, n_] :=
  Integrate[phi1[x, m, l], {x, 0, 1 / (Sqrt[1 + 4 / n^2] + 2 / n)^2}];
p = 0.5;
mmax = 12;
lmax = 10;
Ntable1 = Table[0, {mmax - 2}, {lmax + 1}];
Do[rhs = (p + 1) / 2 * integral[m, l];
Ntable1[[m - 2, l + 1]] = smallestN[Re[integraln[m, l, #]] > rhs &, 3],
{m, 3, mmax}, {l, 0, lmax}];
Print[TableForm[Ntable1]];
```

```
62    101    140    179    218    257    297    336    375    415    454
14    21     28     35     42     49     56     63     70     77     84
8     12     15     18     22     25     28     32     35     38     42
6     8      11     13     15     17     19     21     23     25     27
5     7      8      10     11     13     14     16     18     19     21
4     6      7      8      9      11     12     13     14     15     17
4     5      6      7      8      9      10     11     12     13     14
4     5      5      6      7      8      9      10     10     11     12
3     4      5      6      6      7      8      9      9      10     11
3     4      5      5      6      7      7      8      8      9      10
```

Table 2.2. The computed table `Ntable1` of integers $N_{1,m,l}^{(0.5)}$ for $m = 3, 4, \ldots, 12$ and $l = 0, 1, \ldots, 10$

m \ l	0	1	2	3	4	5	6	7	8	9	10
3	62	101	140	179	218	257	297	336	375	415	454
4	14	21	28	35	42	49	56	63	70	77	84
5	8	12	15	18	22	25	28	32	35	38	42
6	6	8	11	13	15	17	19	21	23	25	27
7	5	7	8	10	11	13	14	16	18	19	21
8	4	6	7	8	9	11	12	13	14	15	17
9	4	5	6	7	8	9	10	11	12	13	14
10	4	5	5	6	7	8	9	10	10	11	12
11	3	4	5	6	6	7	8	9	9	10	11
12	3	4	5	5	6	7	7	8	8	9	10

2.3. Computing the integers $N_{1,m,l}^{(0.9)}$

The following code constructs and prints a table `Ntable1` of integers $N_{1,m,l}^{(0.9)}$ for $m = 3, 4, \ldots, 12$ and $l = 0, 1, \ldots, 10$.

```
phi1[x_, m_, l_] := x^(l / 2) * (1 - x)^(m / 2 - 2);
integral[m_, l_] := Integrate[phi1[x, m, l], {x, 0, 1}];
integraln[m_, l_, n_] :=
  Integrate[phi1[x, m, l], {x, 0, 1 / (Sqrt[1 + 4 / n^2] + 2 / n)^2}];
p = 0.9;
mmax = 12;
lmax = 10;
Ntable1 = Table[0, {mmax - 2}, {lmax + 1}];
Do[rhs = (p + 1) / 2 * integral[m, l];
Ntable1[[m - 2, l + 1]] = smallestN[Re[integraln[m, l, #]] > rhs &, 3],
{m, 3, mmax}, {l, 0, lmax} ];
Print[TableForm[Ntable1]];
```

```
1598    2591    3596    4606    5618    6632    7647    8662    9678    10 694    11 710
78      117     156     195     234     273     312     351     390     429       468
28      40      51      63      74      85      97      108     120     131       142
16      22      28      34      39      45      51      56      62      68        73
12      15      19      23      26      30      33      37      40      44        47
9       12      14      17      20      22      25      27      29      32        34
8       10      12      14      16      18      19      21      23      25        27
7       8       10      12      13      15      16      18      19      21        22
6       7       9       10      11      13      14      15      16      18        19
5       7       8       9       10      11      12      13      14      15        16
```

Table 2.3. The computed table `Ntable1` of integers $N_{1,m,l}^{(0.9)}$ for $m = 3, 4, \ldots, 12$ and $l = 0, 1, \ldots, 10$

m \ l	0	1	2	3	4	5	6	7	8	9	10
3	1598	2591	3596	4606	5618	6632	7647	8662	9678	10694	11710
4	78	117	156	195	234	273	312	351	390	429	468
5	28	40	51	63	74	85	97	108	120	131	142
6	16	22	28	34	39	45	51	56	62	68	73
7	12	15	19	23	26	30	33	37	40	44	47
8	9	12	14	17	20	22	25	27	29	32	34
9	8	10	12	14	16	18	19	21	23	25	27
10	7	8	10	12	13	15	16	18	19	21	22
11	6	7	9	10	11	13	14	15	16	18	19
12	5	7	8	9	10	11	12	13	14	15	16

3. Computing the integers $N_{2,m,l}^{(p)}$

Given $p \in [0, 1[$, the code below constructs and prints a table `Ntable2` of integers $N_{2,m,l}^{(p)}$ for $l = 0, 1, \ldots, l_{\max}$ and $m = 5, 6, \ldots, m_{\max}$. The three occurrences of the symbol ⌘ in the code should be replaced with the actual values of p, $l_{\max}$ and $m_{\max}$.

```
phi2[x_, y_, m_, l_] := (x y)^(l / 2) * ((1 - x) (1 - y))^(m / 2 - 3) * (x - y)
integral2[m_, l_] := Integrate[phi2[x, y, m, l], {x, 0, 1}, {y, 0, x}]
integral2n[m_, l_, n_] :=
 Integrate[phi2[x, y, m, l], {x, 0, 1 / (Sqrt[1 + 8 / n^2] + Sqrt[8] / n)^2}, {y, 0, x}]
p = ⌘;
mmax = ⌘;
lmax = ⌘;
Ntable2 = Table[0, {mmax - 4}, {lmax + 1}];
Do[rhs = (p + 1) / 2 * integral2[m, l];
Ntable2[[m - 4, l + 1]] = smallestN[Re[integral2n[m, l, #]] > rhs &, 3],
{m, 5, mmax}, {l, 0, lmax}];
Print[TableForm[Ntable2]];
```

3.1. Computing the integers $N_{2,m,l}^{(0)}$

The code below constructs and prints a table `Ntable2` of integers $N_{2,m,l}^{(0)}$ for $m = 5, 6, \ldots, 14$ and $l = 0, 1, \ldots, 10$.

```
phi2[x_, y_, m_, l_] := (x y)^(l / 2) * ((1 - x) (1 - y))^(m / 2 - 3) * (x - y)
integral2[m_, l_] := Integrate[phi2[x, y, m, l], {x, 0, 1}, {y, 0, x}]
integral2n[m_, l_, n_] :=
 Integrate[phi2[x, y, m, l], {x, 0, 1 / (Sqrt[1 + 8 / n^2] + Sqrt[8] / n)^2}, {y, 0, x}]
p = 0;
mmax = 14;
lmax = 10;
Ntable2 = Table[0, {mmax - 4}, {lmax + 1}];
Do[rhs = (p + 1) / 2 * integral2[m, l];
Ntable2[[m - 4, l + 1]] = smallestN[Re[integral2n[m, l, #]] > rhs &, 3],
{m, 5, mmax}, {l, 0, lmax}];
Print[TableForm[Ntable2]];
```

```
77    107   137   167   197   227   257   287   317   347   377
25    33    41    49    58    66    74    82    90    98    107
15    20    24    28    33    37    41    46    50    54    59
11    14    17    20    23    26    29    32    34    37    40
9     11    14    16    18    20    22    24    26    29    31
8     10    11    13    15    17    18    20    22    23    25
7     8     10    11    13    14    16    17    18    20    21
6     8     9     10    11    12    14    15    16    17    18
6     7     8     9     10    11    12    13    14    15    16
5     6     7     8     9     10    11    12    13    14    15
```

Table 3.1. The computed table `Ntable2` of integers $N_{2,m,l}^{(0)}$ for $m = 5, 6, \ldots, 14$ and $l = 0, 1, \ldots, 10$

m \ l	0	1	2	3	4	5	6	7	8	9	10
5	77	107	137	167	197	227	257	287	317	347	377
6	25	33	41	49	58	66	74	82	90	98	107
7	15	20	24	28	33	37	41	46	50	54	59
8	11	14	17	20	23	26	29	32	34	37	40
9	9	11	14	16	18	20	22	24	26	29	31
10	8	10	11	13	15	17	18	20	22	23	25
11	7	8	10	11	13	14	16	17	18	20	21
12	6	8	9	10	11	12	14	15	16	17	18
13	6	7	8	9	10	11	12	13	14	15	16
14	5	6	7	8	9	10	11	12	13	14	15

3.2. Computing the integers $N_{2,m,l}^{(0.5)}$

The following code constructs and prints a table `Ntable2` of integers $N_{2,m,l}^{(0.5)}$ for $m = 5, 6, \ldots, 14$ and $l = 0, 1, \ldots, 10$.

```
phi2[x_, y_, m_, l_] := (x y)^(l / 2) * ((1 - x) (1 - y))^(m / 2 - 3) * (x - y)
integral2[m_, l_] := Integrate[phi2[x, y, m, l], {x, 0, 1}, {y, 0, x}]
integral2n[m_, l_, n_] :=
 Integrate[phi2[x, y, m, l], {x, 0, 1 / (Sqrt[1 + 8 / n^2] + Sqrt[8] / n)^2}, {y, 0, x}]
p = 0.5;
mmax = 14;
lmax = 10;
Ntable2 = Table[0, {mmax - 4}, {lmax + 1}];
Do[rhs = (p + 1) / 2 * integral2[m, l];
Ntable2[[m - 4, l + 1]] = smallestN[Re[integral2n[m, l, #]] > rhs &, 3],
{m, 5, mmax}, {l, 0, lmax}];
Print[TableForm[Ntable2]];
```

```
349   484   620   756   892   1028  1165  1301  1438  1574  1711
59    79    99    118   138   158   177   197   217   236   256
29    38    46    54    63    71    80    88    96    105   113
19    24    29    34    39    44    49    54    59    64    69
15    18    22    25    29    32    35    39    42    46    49
12    15    17    20    22    25    28    30    33    35    38
10    12    14    17    19    21    23    25    27    29    31
9     11    13    14    16    18    19    21    23    24    26
8     10    11    13    14    16    17    18    20    21    23
7     9     10    11    13    14    15    16    18    19    20
```

Table 3.2. The computed table `Ntable2` of integers $N_{2,m,l}^{(0.5)}$ for $m = 5, 6, \ldots, 14$ and $l = 0, 1, \ldots, 10$

m \ l	0	1	2	3	4	5	6	7	8	9	10
5	349	484	620	756	892	1028	1165	1301	1438	1574	1711
6	59	79	99	118	138	158	177	197	217	236	256
7	29	38	46	54	63	71	80	88	96	105	113
8	19	24	29	34	39	44	49	54	59	64	69
9	15	18	22	25	29	32	35	39	42	46	49
10	12	15	17	20	22	25	28	30	33	35	38
11	10	12	14	17	19	21	23	25	27	29	31
12	9	11	13	14	16	18	19	21	23	24	26
13	8	10	11	13	14	16	17	18	20	21	23
14	7	9	10	11	13	14	15	16	18	19	20

3.3. Computing the integers $N_{2,m,l}^{(0.9)}$

The following code constructs and prints a table `Ntable2` of integers $N_{2,m,l}^{(0.9)}$ for $m = 5, 6, \ldots, 14$ and $l = 0, 1, \ldots, 10$.

```
phi2[x_, y_, m_, l_] := (x y)^(l / 2) * ((1 - x) (1 - y))^(m / 2 - 3) * (x - y)
integral2[m_, l_] := Integrate[phi2[x, y, m, l], {x, 0, 1}, {y, 0, x}]
integral2n[m_, l_, n_] :=
 Integrate[phi2[x, y, m, l], {x, 0, 1 / (Sqrt[1 + 8 / n^2] + Sqrt[8] / n)^2}, {y, 0, x}]
p = 0.9;
mmax = 14;
lmax = 10;
Ntable2 = Table[0, {mmax - 4}, {lmax + 1}];
Do[rhs = (p + 1) / 2 * integral2[m, l];
Ntable2[[m - 4, l + 1]] = smallestN[Re[integral2n[m, l, #]] > rhs &, 3],
{m, 5, mmax}, {l, 0, lmax}];
Print[TableForm[Ntable2]];
```

```
9037    12 543  16 066  19 598  23 135  26 674  30 216  33 760  37 304  40 849  44 395
331     442     552     662     772     883     993     1103    1214    1324    1434
99      128     157     185     214     243     271     300     329     357     386
51      65      78      91      104     118     131     144     157     170     184
33      41      49      57      65      73      81      89      96      104     112
25      30      36      41      46      52      57      62      67      73      78
20      24      28      32      36      40      43      47      51      55      59
16      20      23      26      29      32      35      38      41      44      47
14      17      19      22      24      27      29      32      34      36      39
12      15      17      19      21      23      25      27      29      31      33
```

Table 3.3. The computed table `Ntable2` of integers $N_{2,m,l}^{(0.9)}$ for $m = 5, 6, \ldots, 14$ and $l = 0, 1, \ldots, 10$

m \ l	0	1	2	3	4	5	6	7	8	9	10
5	9037	12543	16066	19598	23135	26674	30216	33760	37304	40849	44395
6	331	442	552	662	772	883	993	1103	1214	1324	1434
7	99	128	157	185	214	243	271	300	329	357	386
8	51	65	78	91	104	118	131	144	157	170	184
9	33	41	49	57	65	73	81	89	96	104	112
10	25	30	36	41	46	52	57	62	67	73	78
11	20	24	28	32	36	40	43	47	51	55	59
12	16	20	23	26	29	32	35	38	41	44	47
13	14	17	19	22	24	27	29	32	34	36	39
14	12	15	17	19	21	23	25	27	29	31	33

4. Computing the integers $N_{m,l}^{(p)}$

Given $p \in [0, 1[$, the code below constructs and prints a table `Ntable` of integers $N_{m,l}^{(p)}$ for $l = 1, 2, \ldots, l_{\max}$ and $m = 3, 4, \ldots, m_{\max}$. The three occurrences of the symbol ⌘ in the code should be replaced with the actual values of p, $l_{\max}$ and $m_{\max}$.

```
p = ⌘;
lmax = ⌘;
mmax = ⌘;
Ntable = Table[0, {mmax - 2}, {lmax}];
Do[
Ntable[[m - 2, l]] = smallestN[
    Gamma[m / 2 - 1, 0, 2 * Pi * l / #^2] < (1 - p) / 2 * Gamma[m / 2 - 1] &, 2 - (-1)^m],
{m, 3, mmax}, {l, 1, lmax}];
Print[TableForm[Ntable]];
```

4.1. Computing the integers $N_{m,l}^{(0)}$

The code below constructs and prints a table `Ntable` of integers $N_{m,l}^{(0)}$ for $m = 3, 4, \ldots, 12$ and $l = 1, 2, \ldots, 10$.

```
p = 0;
lmax = 10;
mmax = 12;
Ntable = Table[0, {mmax - 2}, {lmax}];
Do[
Ntable[[m - 2, l]] = smallestN[
    Gamma[m / 2 - 1, 0, 2 * Pi * l / #^2] < (1 - p) / 2 * Gamma[m / 2 - 1] &, 2 - (-1)^m],
{m, 3, mmax}, {l, 1, lmax}];
Print[TableForm[Ntable]];
```

```
6   8   10  11  12  13  14  15  16  17
4   5   6   7   7   8   8   9   10  10
3   4   4   5   6   6   7   7   7   8
2   3   4   4   5   5   6   6   6   7
3   3   3   4   4   5   5   5   6   6
2   3   3   4   4   4   5   5   5   5
3   3   3   3   4   4   4   4   5   5
2   2   3   3   3   4   4   4   4   5
3   3   3   3   3   4   4   4   4   4
2   2   3   3   3   3   4   4   4   4
```

Table 4.1. The computed table `Ntable` of integers $N_{m,l}^{(0)}$ for $m = 3, 4, \ldots, 12$ and $l = 1, 2, \ldots, 10$

m \ l	1	2	3	4	5	6	7	8	9	10
3	6	8	10	11	12	13	14	15	16	17
4	4	5	6	7	7	8	8	9	10	10
5	3	4	4	5	6	6	7	7	7	8
6	2	3	4	4	5	5	6	6	6	7
7	3	3	3	4	4	5	5	5	6	6
8	2	3	3	4	4	4	5	5	5	5
9	3	3	3	3	4	4	4	4	5	5
10	2	2	3	3	3	4	4	4	4	5
11	3	3	3	3	3	4	4	4	4	4
12	2	2	3	3	3	3	4	4	4	4

4.2. Computing the integers $N_{m,l}^{(0.5)}$

The following code constructs and prints a table `Ntable` of integers $N_{m,l}^{(0.5)}$ for $m = 3, 4, \ldots, 12$ and $l = 1, 2, \ldots, 10$.

```
p = 0.5;
lmax = 10;
mmax = 12;
Ntable = Table[0, {mmax - 2}, {lmax}];
Do[
Ntable[[m - 2, l]] = smallestN[
    Gamma[m / 2 - 1, 0, 2 * Pi * l / #^2] < (1 - p) / 2 * Gamma[m / 2 - 1] &, 2 - (-1)^m],
{m, 3, mmax}, {l, 1, lmax}];
Print[TableForm[Ntable]];
```

```
12    16    20    23    25    28    30    32    34    36
5     7     9     10    11    12    13    14    15    15
4     5     6     7     8     8     9     10    10    11
3     4     5     6     6     7     7     8     8     9
3     4     4     5     5     6     6     7     7     7
2     3     4     4     5     5     6     6     6     7
3     3     3     4     4     5     5     5     6     6
2     3     3     4     4     4     5     5     5     5
3     3     3     3     4     4     4     5     5     5
2     2     3     3     4     4     4     4     5     5
```

Table 4.2. The computed table `Ntable` of integers $N_{m,l}^{(0.5)}$ for $m = 3, 4, \ldots, 12$ and $l = 1, 2, \ldots, 10$

m \ l	1	2	3	4	5	6	7	8	9	10
3	12	16	20	23	25	28	30	32	34	36
4	5	7	9	10	11	12	13	14	15	15
5	4	5	6	7	8	8	9	10	10	11
6	3	4	5	6	6	7	7	8	8	9
7	3	4	4	5	5	6	6	7	7	7
8	2	3	4	4	5	5	6	6	6	7
9	3	3	3	4	4	5	5	5	6	6
10	2	3	3	4	4	4	5	5	5	5
11	3	3	3	3	4	4	4	5	5	5
12	2	2	3	3	4	4	4	4	5	5

4.3. Computing the integers $N_{m,l}^{(0.9)}$

The following code constructs and prints a table `Ntable` of integers $N_{m,l}^{(0.9)}$ for $m = 3, 4, \ldots, 12$ and $l = 1, 2, \ldots, 10$.

```
p = 0.9;
lmax = 10;
mmax = 12;
Ntable = Table[0, {mmax - 2}, {lmax}];
Do[
Ntable[[m - 2, l]] = smallestN[
    Gamma[m / 2 - 1, 0, 2 * Pi * l / #^2] < (1 - p) / 2 * Gamma[m / 2 - 1] &, 2 - (-1)^m],
{m, 3, mmax}, {l, 1, lmax}];
Print[TableForm[Ntable]];
```

```
57    80    98    114   127   139   150   160   170   179
12    16    20    23    25    28    30    32    34    35
6     9     11    12    14    15    16    17    18    19
5     6     8     9     10    11    12    12    13    14
4     5     6     7     8     9     9     10    10    11
3     4     5     6     7     7     8     8     9     9
3     4     5     5     6     6     7     7     8     8
3     4     4     5     5     6     6     7     7     7
3     3     4     4     5     5     6     6     6     7
2     3     4     4     4     5     5     6     6     6
```

Table 4.3. The computed table `Ntable` of integers $N_{m,l}^{(0.9)}$ for $m = 3, 4, \ldots, 12$ and $l = 1, 2, \ldots, 10$

m \ l	1	2	3	4	5	6	7	8	9	10
3	57	80	98	114	127	139	150	160	170	179
4	12	16	20	23	25	28	30	32	34	35
5	6	9	11	12	14	15	16	17	18	19
6	5	6	8	9	10	11	12	12	13	14
7	4	5	6	7	8	9	9	10	10	11
8	3	4	5	6	7	7	8	8	9	9
9	3	4	5	5	6	6	7	7	8	8
10	3	4	4	5	5	6	6	7	7	7
11	3	3	4	4	5	5	6	6	6	7
12	2	3	4	4	4	5	5	6	6	6